Instructions

Intro to Tidyverse: Tidying and Plotting Basics

GOVT16/QSS17: Data Visualization (Prof. Cooper)

Week 2

- Date: April 7th - April 14th (Due April 14th, 11:59 PM)

Preparation

1. During this term, you will use a specific folder within your computer – your "working directory" – where you will save your R scripts, data, figures, and other documents. To make this folder, open RStudio and go to "File" > "New Project". Then, choose "New Directory", and make a new "project" (i.e., a new directory) somewhere on your computer. The project name, which is the name of your working directory, can be anything. Check that the new folder is generated with a new `.Rproj` file. As long as all of your files for this course are saved in this working directory, you need to make this `.Rproj` file once. When you work on your own project after this course, however, you should specify a different working directory with a new `.Rproj` file for each project.

What should you do in Week 2?

- In RStudio, choose "File" > "New File" > "R Script". Then, save the file as "Week02_script.R".
- Work on Exercises below.
- You should upload your R script by April 14th (Tuesday) 11:59 pm ("Assignments" > "Week 2 Assignment").

Notes:

- At this stage I encourage you to write reasonably extensive comments in your own version of your scripts so that you remember what it is you did and why

you did it. Scripts to be read by others, under normal circumstances, should not be quite so heavyhanded with comments.

- Never use `T` and `F` as synonyms for `TRUE` and `FALSE`, because `T` and `F` can be redefined and could cause confusion and error. See `https://www.r-bloggers.com/r-tip-avoid-using-t-and-f-as-synonyms-for-true-and-false/`

- Each time I ask you to create a new object, please follow the creation of that object by printing the object. Under normal circumstances you would not print out every object in your R script, whether using proper workflow on your own or sharing scripts with other researchers and collaborators. It will be useful for all of us *in the context of this course*. You can check your work constantly by seeing the output and I can grade more efficiently by seeing the output.

- If I use a particular verb or reference a particular operator or function that you do not immediately recognize, there are many resources out there to help you to discover the solution. The process of search and discovery for basic coding solutions is a part of programming, and **R** has many online resources for you. Google is your friend, along with a very large number of sources for **R** support.

Exercises

Set up the top of your R script with your name, the name of the course, the term, and the date using comments. There are different style guides for comments. For your homeworks, please use `#` (followed by one space) to add a comment to your code. Begin with `##` (followed by one space) for entirely commented lines.

```
> ## Data Visualization (GOVT16-QSS17) Spring 2026
> ## Intro to the Tidyverse
> ##
> ## Name: YOUR NAME GOES HERE
> ## Date: April 7th - April 14th, 2026
```

1. **Filter**

   (a) Load the `tidyverse` package (Actually, it's a package of packages. You'll see.) with the `library` command. If you do not already have the package installed, you can do so from the `Tools` portion of the `RStudio` menu or with the `install.packages()` command. It's basically free points, here.

   (b) Load the `swiss` dataset with the `data` command. Look at the `head` of the dataset. These data represent a few key socioeconomic indicators for some of the provinces of Switzerland in the late-nineteenth century. If you want to see more about the data, use the question mark helper function `?swiss`.

   (c) Turn the dataset into a `tibble` if it is not already one. (I will never ask you to do this again. Tibbles are "teribble.")

   (d) Filter the dataset for provinces that have less that 50% male involvement in `agriculture`. Use the `pipe` operator when doing so, please. Recall: the pipe operator is a part of the `tidyverse` through the `magrittr` package.

   (e) Filter the `swiss` dataset two ways this time. Filter the data for provinces above 50% `Catholic` and for provinces and below 70% `fertility`. You should be able to do this in one "verb" command.

2. **Arrange**

   (a) Take the `swiss` dataset and arrange it by `Infant.Mortality`.

   (b) Arrange the `swiss` dataset by `Catholic` in *descending* order.

   (c) Let's do some combination. Take the `swiss` dataset, filter the data for provinces with less than 50% `Catholics` and arrange by `Fertility`.

3. **Mutate**

   (a) Download the IMDB movie dataset onto your computer and name it `movies`. It is a large, ugly dataset with over 5,000 movies. You will have to work with datasets like this one in your life, probably often. Better to get used to it now. To import the data into **R** I recommend

you try the `read_csv` function from the `readr` package. You can either code the entire path directory of the file in quotes or you can stuff a `file.choose()` function inside the `read_csv` function, *exactly* like this: `read_csv(file.choose())`.

(b) Explore the `movie` dataset by looking at its structure via `glimpse` or `str`.

(c) The budgets of movies are measured in individual dollars. Add a new variable to the dataset `budget2` with the `mutate` function. Make it so that `budget2` is measured in *millions of dollars*.

(d) Who can count in minutes? Seriously. Let's add a `length` variable that is properly measured in hours, which is how normal people measure the length of movies.

(e) Subset the `movie` dataset from "Spectre" to "Skyfall." Use the titles in `movie_title` to do this, not row and column numbers like we did in the first assignment. You may need to combine some functions/commands in order to do this. This will require a function that works with characters/strings (maybe try the `stringr` package!). *Note: warnings are not the same things as errors.*

4. **Plots & `ggplot2`**

(a) Read in the United Nations World Income Inequality Database (WIID). These are fresh data, updated and published in May 2021. You may call the dataset whatever you like, but I suggest something like `undat` or `wiid`. In order to do this, you'll need to install another package, `readxl`, using the `read_excel()` function.

(b) Show me the `head` and `tail` of the dataset. There are a lot of variables in this data set, so `glimpse` it as well.

(c) Filter the dataset for just North American countries (make sure to include Mexico!). Produce a scatterplot of Gini coefficient, `gini`, over time (use the `year` variable on the x-axis). Map `country` onto color (this is another way of saying add a color aesthetic). There is a little bit of overplotting, but we can see the basic trends pretty well.

(d) Repeat the problem above, but this time make a line plot instead. Note how much movement is happening within the country lines; they should look a bit messy, ugly, and inappropriate. Investigate the data a bit to try to figure out why and tell me in a comment.

5. **Basic statistics and summaries with `summarize`**

(a) Use the `pipe` operator and `summarize` the Gini index `gini`, the first quintile `q1` and the fifth quintile `q5`, giving me the median of each of these using a single `summarize` function. If you save the output as an object, make sure you also print it so that I can see it. *You do not need any grouping for this problem.*

(b) I'm curious. For the year 2000, what are the highest values for the Gini index (`gini`), 1st decile (`d1`), and 10th decile (`d10`) values reported for Africa? (The deciles represent percentages of countries' population and their income held, so `d1` reflects the percent of a country's income held by the poorest 10% of the population).

(c) Filter the `WIID` for the United States in the year 2000 and show the Gini index value(s) *without* summarizing them. This should be a little instructive about a plot you might have tried earlier. Recall the problem where you created a scatterplot and a line plot for North American countries? This problem should help you understand why they were overplotting.

(d) Filter the `WIID` for African countries, reporting the means, medians, and the standard deviations of Gini coefficients *for Africa's UN sub-regions*. *Hint*: There should be some grouping going on here, and the result should be just a few rows with four columns.

6. **More Plots**

(a) Group the `WIID` data by UN region and year, and then make separate scatterplots (one plot environment, but faceted) of each region's *mean* Gini index values. Clean up your labels (that goes for all the plots). The x-axis, y-axis, title(s), and legend(s) can all be retitled inside one line under most circumstances.

(b) Group the `WIID` data by OECD status and year, plotting scatter plots for median Gini index values by OECD status. Add a smoother line with `geom_smooth`.

(c) Create another bar plot of median Gini index values for each UN region by year. Filter for years after 1945. Add some color for regions and facet the plots by UN region.

(d) Create a histogram of the percentage of income held by the top decile `d10` in countries across the world. Please add a fill aesthetic separating the histogram by UN regions. (Remind yourself why I would ask for "fill" as opposed to "color")

(e) Note how the y-axis on the previous histogram was a count. We can change a histogram to behave more like a density plot, where the y-axis accounts for the proportion or density of each region's values. Take the previous histogram code and adapt it so that regions overlap on the plot (the default `position` argument is a stacked histogram). Add another aesthetic, a `y` aesthetic, that changes the underlying statistic performed to a density and not a count. A quick Google should help you locate this. Finally, set the `alpha` transparency argument to about 0.6 to add some transparency, please.

(f) Create some box plots of the lowest quintile (`q1`) values of income split into groups by UN region. This is another way of seeing income inequality beyond just the Gini scores.

(g) Finally, I want to see if income inequality within countries varies by income group across the UN regions. Make a series of box plots (using a facet function) that show Gini scores broken down by `incomegroup` and faceted by UN region. Make sure the `incomegroup` categories are in the right order. *Hint*: you may need to check how to change the size or angle of axis labels to be able to see them all. Alternatively, you could `flip` the axes.